Software Test Plan (STP) - Hospital Management System

Project: Hospital Management System
Version: 1.0
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1. Introduction

Purpose: This document defines how Hospital Management System version 1.0 will be verified against the approved SRS. It establishes the scope, strategy, environment, responsibilities, schedule, entry and exit criteria, traceability, and reporting controls for the project team and evaluator.

Scope: Testing covers the C/C++ application and its local persistent storage, including authentication, patient and doctor records, appointments, admissions and room allocation, medical records, billing and payments, reports, validation, backup, persistence, and safe exit. Testing includes positive, negative, boundary, integration, recovery, usability, performance, security, portability, and acceptance checks.

References: Hospital Management System Software Requirements Specification; agreed C/C++ design and source code; compiler/build instructions; approved requirements traceability matrix.

Definitions: HMS (Hospital Management System), SRS (Software Requirements Specification), CRUD (Create, Read, Update, Delete), RTM (Requirements Traceability Matrix), QA (Quality Assurance), UAT (User Acceptance Testing).

2. Test Items

- User authentication and role access
- Patient registration, update, search, and status management
- Doctor and department management
- Appointment scheduling, rescheduling, cancellation, and conflict validation
- Admission, room or bed allocation, occupancy, and discharge
- Medical records, diagnosis, treatment, and prescription information
- Billing, payments, balances, and billing summaries
- Administrative reports and record retrieval
- Input validation, file persistence, backup, and safe exit

3. Features to be Tested

Features mapped to SRS requirement IDs:
- FR-01: User login, authentication, and role-appropriate access
- FR-02: Patient registration and unique record creation
- FR-03: Permitted patient-record updates with a stable patient ID
- FR-04: Patient search by ID, name, contact number, and supported fields
- FR-05: Doctor information, specialization, department, contact, and availability
- FR-06: Appointment creation for valid patients and doctors
- FR-07: Appointment rescheduling, cancellation, status, and conflict handling
- FR-08: Admission details and attending-doctor linkage
- FR-09: Available room or bed display and allocation
- FR-10: Diagnosis, symptoms, treatment notes, prescriptions, and visit history
- FR-11: Discharge date, summary, room release, and final status
- FR-12: Bill creation and charge or total calculation
- FR-13: Payment recording, status, and outstanding balance
- FR-14: Patient, appointment, admission, occupancy, and billing reports
- FR-15: Mandatory-field, identifier, date, numeric, boundary, and duplicate validation
- FR-16: Persistent save, load, and restart recovery
- FR-17: Data backup or export
- FR-18: Graceful exit and safe save
- NFR: Response time, usability, reliability, security and privacy, maintainability evidence, portability, and successful compilation

4. Features Not to be Tested

- Real hospital devices and external healthcare-system integrations
- External insurance, laboratory, pharmacy, and payment-gateway processing
- Online appointment booking, email or SMS delivery, cloud deployment, and multi-user networking
- Database products, GUI or web or mobile clients, and analytics listed only as future enhancements
- Clinical correctness or replacement of professional medical judgment
- Production-scale load, regulatory certification, and security assessment beyond the academic demonstration scope

5. Test Approach / Strategy

Levels:
- Unit testing of validation, ID generation, calculations, file operations, and module functions
- Integration testing across patient, doctor, appointment, admission, medical record, billing, reporting, and storage modules
- System testing of complete role-based workflows from login through save, restart, reporting, and exit
- User acceptance testing against SRS use cases UC-01 to UC-06 and acceptance criteria

Types and techniques:

- Functional, regression, smoke, negative, boundary-value, equivalence-partition, recovery, usability, security, portability, and performance testing
- Manual scenario execution supported by repeatable C/C++ unit tests and scripted test-data setup where practical

Entry Criteria: A compilable tagged build is available; requirements and build instructions are baselined; the test environment and seeded test data are ready; known limitations are documented.
Exit Criteria: All planned high-priority tests and all SRS acceptance workflows are executed; all FR-01 to FR-18 requirements have test evidence; no open critical or high-severity defect blocks a core workflow; failed tests have accepted disposition; the final summary and RTM are complete.

5.1 Security and Privacy Validation

- Verify valid and invalid login behavior and deny protected functions to unauthorized roles
- Confirm administrative, receptionist, doctor, and billing functions expose only permitted operations and data
- Verify passwords are masked in the interface and are not written to logs or ordinary report output
- Test malformed, oversized, empty, duplicate, and boundary input without crashes or unsafe records
- Verify sensitive patient and medical information is not unnecessarily displayed across roles
- Review local data-file access, backup copies, and failure messages for accidental disclosure or corruption
- Confirm no external network communication is required by the initial release

6. Test Environment

Hardware: Standard dual-core computer or better, at least 4 GB RAM, keyboard, display, and at least 500 MB free storage.
Software: Windows 10 or 11 and a compatible Linux environment where available; GCC/G++ or MinGW or equivalent C/C++ compiler; project build tools and standard library.
Configuration: Clean build directory, isolated test-data folder, known seed files, writable and read-only storage variants, and backup and restore location.
Tools: Compiler diagnostics, C/C++ unit-test harness where implemented, command-line or shell scripts for repeatable runs, debugger or memory checker where available, and the project issue tracker or defect log.
Test Data: Synthetic patients, doctors, appointments, admissions, rooms, medical records, bills, and payments covering valid, invalid, duplicate, boundary, missing, and conflicting values. No real patient data will be used.

7. Test Schedule

Milestones (relative to the approved project start):
- Week 1: Baseline the plan and RTM; prepare the environment, data, and detailed tests

- Week 2: Execute unit, integration, validation, storage, system, security, usability, and portability tests
- Week 3, Days 1-2: Verify fixes and run regression, performance, and recovery checks
- Week 3, Days 3-4: UAT against UC-01 to UC-06 and acceptance criteria
- Week 3, Day 5: Complete RTM, defect review, final test summary, and approval

8. Test Deliverables

Planned deliverables:
- Approved Software Test Plan
- Requirements Traceability Matrix covering FR-01 to FR-18 and non-functional requirements
- Detailed manual and automated test cases with expected results
- Test data and repeatable setup or reset instructions
- Test scripts or unit-test source where implemented
- Build and environment configuration record
- Test execution log and evidence
- Defect reports with severity, status, and retest result
- Final Test Summary Report and approval record

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead and Test Planner	Pratham P Mudakavi	Own the plan and RTM; coordinate execution, reporting, and final quality review
Test Engineer and Environment Owner	Pratham P Shetty	Design and execute tests; prepare data and environments; record evidence and defects
Development Lead	Prathik M Nambiar	Provide builds and technical support; triage and fix defects; support unit and integration testing
Requirements and Acceptance Owner	Pratik Patil	Confirm SRS coverage; review acceptance results, unresolved risks, and release readiness

10. Risks and Mitigation

Risk	Mitigation
Late or unstable test build	Use early smoke builds and replan by SRS priority
Corrupted or inconsistent persistent files	Use seeded data, clean backups, and save or restart checks
Incomplete role and privacy controls	Run negative role tests and review sensitive fields
Limited time or environment availability	Prioritize high requirements and keep a reproducible setup
Requirement or data-format changes	Update the RTM and affected cases, then run focused regression

11. Assumptions & Dependencies

- The SRS version supplied with this plan is the requirements baseline for version 1.0
- The application is a standalone C/C++ academic mini project using local persistent files
- Test builds include the source code, build instructions, and all required sample data formats
- The team can create synthetic records and reset or restore the test-data directory
- Windows or Linux compiler access is available; cross-platform execution depends on the environments available to the team
- No external insurance, laboratory, payment, device, cloud, or network service is required
- Requirement changes will be reflected in the RTM and affected tests before sign-off

12. Suspension & Resumption Criteria

Suspend testing if:

- The build does not compile or start in the agreed environment
- Login or a shared storage defect prevents more than 30% of planned cases
- Test data is corrupted and cannot be restored, or results cannot be reproduced
- A critical defect risks loss or disclosure of test records
- Requirements or interfaces change without an updated baseline

Resume testing when:

- A corrected build passes compilation and smoke tests
- Blocking defects are fixed or formally deferred with an accepted workaround
- The environment and clean test data are restored and verified
- Changed requirements and affected test cases are updated in the RTM

13. Test Case Management & Traceability

The RTM is the controlling record that maps every SRS requirement to one or more test cases, execution evidence, defects, and final status. Test-case IDs use the module prefix below and remain stable across reruns.

Requirement-to-test mapping:
- FR-01 -> TC-AUTH-01 to TC-AUTH-06
- FR-02 -> TC-PAT-01 to TC-PAT-05
- FR-03 -> TC-PAT-06 to TC-PAT-09
- FR-04 -> TC-PAT-10 to TC-PAT-14
- FR-05 -> TC-DOC-01 to TC-DOC-07
- FR-06 -> TC-APT-01 to TC-APT-06
- FR-07 -> TC-APT-07 to TC-APT-12
- FR-08 -> TC-ADM-01 to TC-ADM-04
- FR-09 -> TC-ADM-05 to TC-ADM-08
- FR-10 -> TC-MED-01 to TC-MED-09
- FR-11 -> TC-ADM-09 to TC-ADM-12
- FR-12 -> TC-BIL-01 to TC-BIL-06
- FR-13 -> TC-BIL-07 to TC-BIL-12
- FR-14 -> TC-RPT-01 to TC-RPT-07
- FR-15 -> TC-VAL-01 to TC-VAL-12
- FR-16 -> TC-STO-01 to TC-STO-07
- FR-17 -> TC-BAK-01 to TC-BAK-04
- FR-18 -> TC-EXIT-01 to TC-EXIT-04
- Performance -> TC-NF-PERF-01 to TC-NF-PERF-04
- Usability -> TC-NF-USA-01 to TC-NF-USA-05
- Reliability -> TC-NF-REL-01 to TC-NF-REL-06
- Security and privacy -> TC-NF-SEC-01 to TC-NF-SEC-07
- Maintainability and portability evidence -> TC-NF-MNT-01 to TC-NF-MNT-03 and TC-NF-PORT-01 to TC-NF-PORT-04
- UC-01 to UC-06 -> TC-UAT-01 to TC-UAT-06

Each test record contains its ID, linked requirement, objective, preconditions, data, steps, expected result, actual result, status, build, executor, date, evidence reference, and defect ID if applicable.

14. Test Metrics & Reporting

Metrics collected:
- Planned, executed, passed, failed, blocked, and not-run test counts and percentages
- Requirement coverage for FR-01 to FR-18 and non-functional requirement categories
- Defect count by severity, status, module, age, reopen rate, and discovery phase
- Pass rate by module and build; regression and retest completion
- Common-operation response time for representative project-scale datasets
- Build compilation, portability, persistence, backup, and restart results

Reporting:
- Execution status at the end of each planned test session
- Immediate escalation of critical or high-severity blocking defects
- Updated defect log and RTM after each test cycle
- Final Test Summary Report with scope, environment, results, unresolved risks, acceptance status, and approval decision

15. Approvals

Role	Name	Signature / Date
QA Lead	Pratham P Mudakavi	
Development Lead	Prathik M Nambiar	
Requirements and Acceptance Owner	Pratik Patil	